

Flow Matching · arXiv 2210.02747

Archaeologist

11+3 · seat 5

CNF $\rightarrow$ FM/CFM $\rightarrow$ DP $\rightarrow \pi_0 \rightarrow$ OpenPI $\rightarrow$ BEHAVIOR

Sep 23 paper club · VLA Architectures

Spine: OT-CFM objective + path design for action statistics

Lipman et al. · Meta AI / Weizmann · pick a path, regress the VF

Intellectual history

Cite hygiene on Lipman vs DP

Lineage · CNF → FM → DP → π_0 → BEHAVIOR

Archaeologist spine · DP likely does NOT cite Lipman · π_0 CLEAR Lipman cite



Diffusion Policy note

Action-space DDPM chunks · conceptual bridge
likely does NOT cite Lipman · cite Chi'23 for DP

π_0 / OpenPI note

CLEAR Lipman cite · OT-style CFM action expert
 $\pi_{0.5}$ / OpenPI keep FM head · BEHAVIOR specializes

Concurrent: Rectified Flow (Liu et al.) / stochastic interpolants (Albergo & Vanden-Eijnden)
Honesty: originality is framing + scale + OT emphasis — not unique loss discovery

One-liner · the club spine



Moral for Sep 23

Next wins = path design for action statistics — not another ImageNet U-Net

Diffusion Policy showed generative chunks · π_0 put OT-CFM in the VLM · BEHAVIOR fixes Σ / N / soft-inpaint

FM = Continuous Normalizing Flow trained by Conditional Flow Matching · OT = straight displacement interpolant

Conditional Flow Matching · the objective card

Thm 2: $\nabla \text{CFM} = \nabla \text{FM}$ · never need the marginal VF

Intractable FM

$$L_{FM} = E_{t, p_t} \|v_t(x) - u_t(x)\|^2$$

needs marginal generating field u_t

hard — mix over all data

Tractable CFM ★

$$L_{CFM} = E_{t, q, x} \|v_t - u_t(x|x_1)\|^2$$

per-sample conditional field

same gradients (Thm 2)

OT-CFM recipe (OpenPI / π_0 action expert)

1. Sample path

$$\begin{aligned} t &\sim U(0,1) \\ \varepsilon &\sim N(0,I), \quad a \sim \text{data} \\ a_t &= (1-t)\varepsilon + t a \end{aligned}$$

2. Target VF

$$\begin{aligned} u &\approx a - \varepsilon \\ &\text{(constant direction)} \\ \sigma_{\min} &\text{softens endpoint} \end{aligned}$$

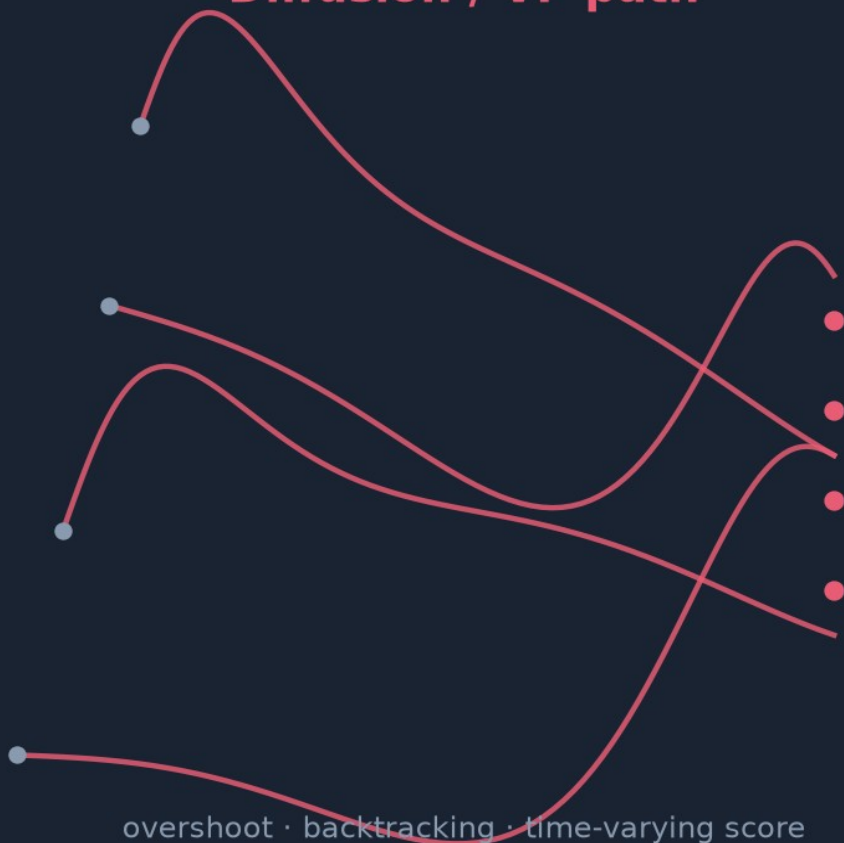
3. Regress

$$\begin{aligned} &\|v_{\theta}(a_t, t | o) - u\|^2 \\ &\text{condition on obs/VLM} \\ &\rightarrow \text{action-chunk expert} \end{aligned}$$

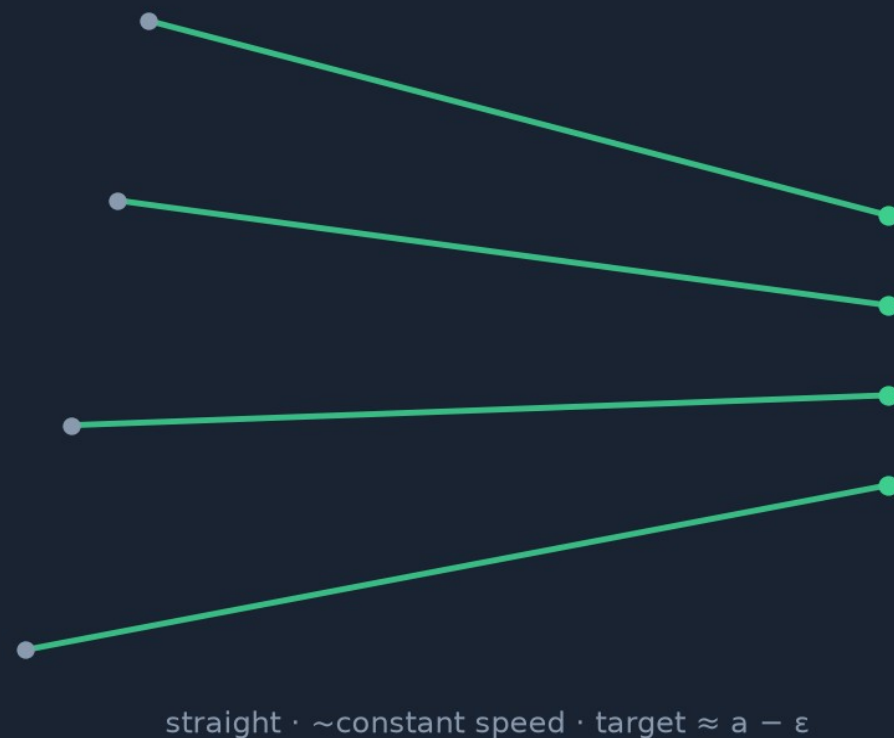
OT straight paths vs diffusion curved paths

Paper Figs 2-3 · constant-direction VF · fewer NFEs for same error

Diffusion / VP path



OT displacement path



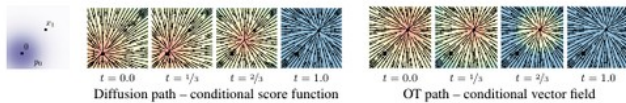


Figure 2: Compared to the diffusion path’s conditional score function, the OT path’s conditional vector field has constant direction in time and is arguably simpler to fit with a parametric model. Note the blue color denotes larger magnitude while red color denotes smaller magnitude.

which, in contrast to the diffusion conditional VF (equation 19), is defined for all $t \in [0, 1]$. The conditional flow that corresponds to $u_t(x|x_1)$ is

$$\psi_t(x) = (1 - (1 - \sigma_{\min})t)x + tx_1, \quad (22)$$

and in this case, the CFM loss (see equations 9, 14) takes the form:

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t,q(x_1),p(x_0)} \left\| v_t(\psi_t(x_0)) - \left(x_1 - (1 - \sigma_{\min})x_0 \right) \right\|^2. \quad (23)$$

Allowing the mean and std to change linearly not only leads to simple and intuitive paths, but it is actually also optimal in the following sense. The conditional flow $\psi_t(x)$ is in fact the Optimal Transport (OT) *displacement map* between the two Gaussians $p_0(x|x_1)$ and $p_1(x|x_1)$. The OT *interpolant*, which is a probability path, is defined to be (see Definition 1.1 in McCann (1997)):

$$p_t = [(1-t)\text{id} + t\psi]_* p_0 \quad (24)$$

where $\psi : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is the OT map pushing p_0 to p_1 , id denotes the identity map, i.e., $\text{id}(x) = x$, and $(1-t)\text{id} + t\psi$ is called the OT displacement map. Example 1.7 in McCann (1997) shows, that in our case of two Gaussians where the first is a standard one, the OT displacement map takes the form of equation 22.

Intuitively, particles under the OT displacement map always move in straight line trajectories and with constant speed. Figure 3 depicts sampling paths for the diffusion and OT conditional VFs. Interestingly, we find that sampling trajectory from diffusion paths can “overshoot” the final sample, resulting in unnecessary backtracking, whilst the OT paths are guaranteed to stay straight.

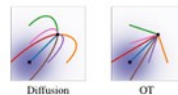


Figure 3: Diffusion and OT trajectories.

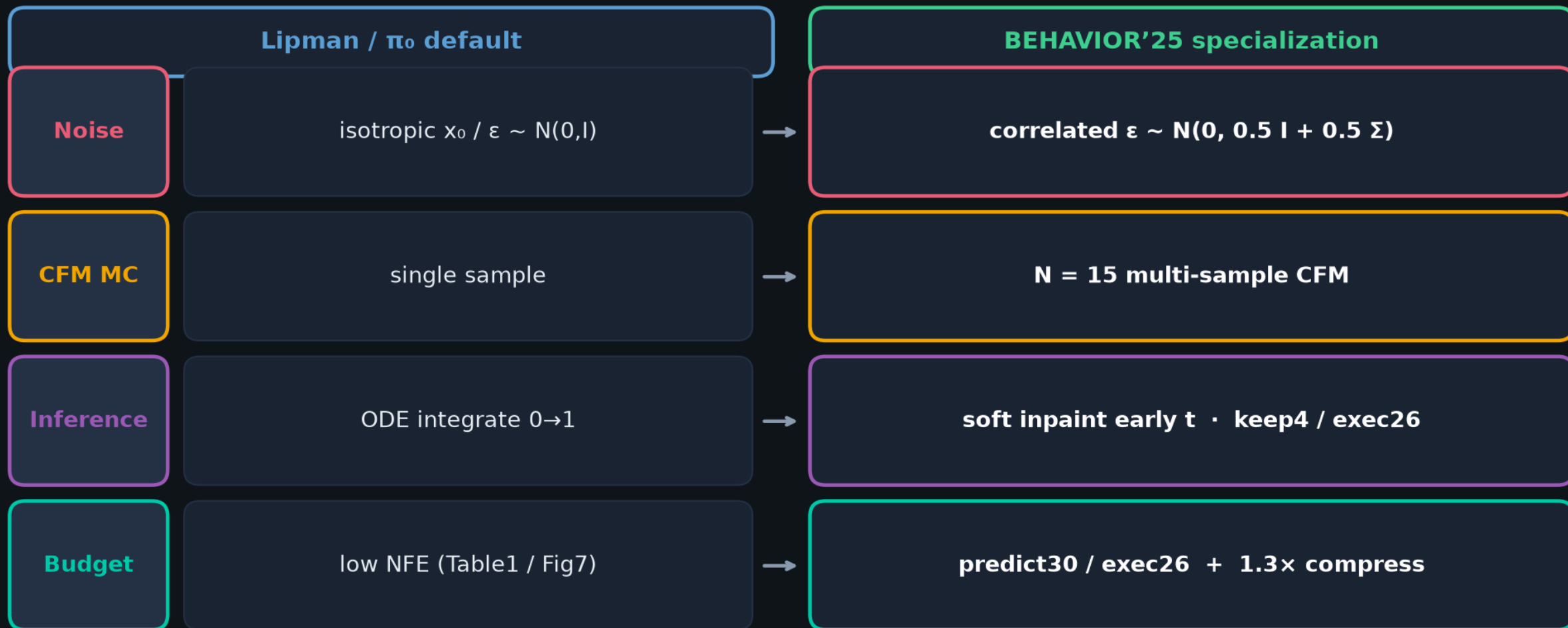
Figure 2 compares the diffusion conditional score function (the regression target in a typical diffusion methods), i.e., $\nabla \log p_t(x|x_1)$ with p_t defined as in equation 18, with the OT conditional VF (equation 21). The start (p_0) and end (p_1) Gaussians are identical in both examples. An interesting observation is that the OT VF has a constant direction in time, which arguably leads to a simpler regression task. This property can also be verified directly from equation 21 as the VF can be written in the form $u_t(x|x_1) = g(t)h(x|x_1)$. Figure 8 in the Appendix shows a visualization of the Diffusion VF. Lastly, we note that although the conditional flow is optimal, this by no means imply that the marginal VF is an optimal transport solution. Nevertheless, we expect the marginal vector field to remain relatively simple.

5 RELATED WORK

Continuous Normalizing Flows were introduced in (Chen et al., 2018) as a continuous-time version of Normalizing Flows (see e.g., Kobayev et al. (2020); Papamakarios et al. (2021) for an overview). Originally, CNFs are trained with the maximum likelihood objective, but this involves expensive ODE simulations for the forward and backward propagation, resulting in high time complexity due to the sequential nature of ODE simulations. Although some works demonstrated the capability of CNF generative models for image synthesis (Grathwohl et al., 2018), scaling up to very high dimensional images is inherently difficult. A number of works attempted to regularize the ODE to be easier to solve, e.g., using augmentation (Dupont et al., 2019), adding regularization terms (Yang & Karniadakis, 2019; Finlay et al., 2020; Onken et al., 2021; Tong et al., 2020; Kelly et al., 2020), or stochastically sampling the integration interval (Du et al., 2022). These works merely aim to regularize the ODE but do not change the fundamental training algorithm.

FM → BEHAVIOR map · what changed for robot stats

Lipman'22 foundation → 2025 BEHAVIOR winners on Pi0.5 / OpenPI



Don't overclaim

DP $\neq$ FM

Diffusion Policy = action-space DDPM. Conceptual bridge only. Likely no Lipman cite.

Σ / N=15 $\neq$ Lipman

Correlated noise + multi-sample CFM are BEHAVIOR specializations, not paper defaults.

IN FID $\neq$ robot KPI

ImageNet tables don't transfer to success rate / Hz / contact richness.

Conditional OT $\neq$ marginal OT

Authors admit marginal VF need not be W_2 -optimal — branding can overclaim.

Concurrent literature

Rectified Flow / interpolants overlap — cite them; claim framing+scale+OT emphasis.

FLOW MATCHING FOR GENERATIVE MODELING

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ABSTRACT

We introduce a new paradigm for generative modeling built on Continuous Normalizing Flows (CNFs), allowing us to train CNFs at unprecedented scale. Specifically, we present the notion of Flow Matching (FM), a simulation-free approach for training CNFs based on regressing vector fields of fixed conditional probability paths. Flow Matching is compatible with a general family of Gaussian probability paths for transforming between noise and data samples—which subsumes existing diffusion paths as specific instances. Interestingly, we find that employing FM with diffusion paths results in a more robust and stable alternative for training diffusion models. Furthermore, Flow Matching opens the door to training CNFs with other, non-diffusion probability paths. An instance of particular interest is using Optimal Transport (OT) displacement interpolation to define the conditional probability paths. These paths are more efficient than diffusion paths, provide faster training and sampling, and result in better generalization. Training CNFs using Flow Matching on ImageNet leads to consistently better performance than alternative diffusion-based methods in terms of both likelihood and sample quality, and allows fast and reliable sample generation using off-the-shelf numerical ODE solvers.

1 INTRODUCTION

Deep generative models are a class of deep learning algorithms aimed at estimating and sampling from an unknown data distribution. The recent influx of amazing advances in generative modeling, *e.g.*, for image generation Ramesh et al. (2022); Rombach et al. (2022), is mostly facilitated by the scalable and relatively stable training of diffusion-based models Ho et al. (2020); Song et al. (2020b). However, the restriction to simple diffusion processes leads to a rather confined space of sampling probability paths, resulting in very long training times and the need to adopt specialized methods (*e.g.*, Song et al. (2020a); Zhang & Chen (2022)) for efficient sampling.

In this work we consider the general and deterministic framework of Continuous Normalizing Flows (CNFs; Chen et al. (2018)). CNFs are capable of modeling arbitrary probability path and are in particular known to encompass the probability paths modeled by diffusion processes (Song et al., 2021). However, aside from diffusion that can be trained efficiently via, *e.g.*, denoising score matching (Vincent, 2011), no scalable CNF training algorithms are known. Indeed, maximum likelihood training (*e.g.*, Grathwohl et al. (2018)) require expensive numerical ODE simulations, while existing simulation-free methods either involve intractable integrals (Rozen et al., 2021) or biased gradients (Ben-Hamu et al., 2022).

The goal of this work is to propose Flow Matching (FM), an efficient simulation-free approach to training CNF models, allowing the adoption of general probability paths to supervise CNF training. Importantly, FM breaks the barriers for scalable CNF training beyond diffusion, and sidesteps the need to reason about diffusion processes to directly work with probability paths.

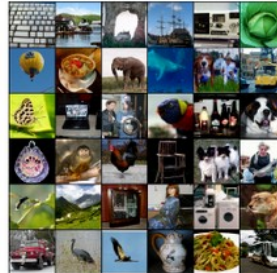


Figure 1: Unconditional ImageNet-128 samples of a CNF trained using Flow Matching with Optimal Transport probability paths.

Visionary arc · one slide



Internalize from THIS paper

OpenPI loss = CFM · OT target $\approx a - \varepsilon$ along $a_t = (1-t)\varepsilon + t a$
Straight paths $\Rightarrow$ fewer NFEs (~ 10 Euler) · Diffusion $\subset$ FM — “back to DDPM” needs a reason

Don't overclaim: DP $\neq$ FM · correlated noise / N=15 $\neq$ Lipman · IN FID $\neq$ robot KPI

Archaeologist · close / Q&A

Habit persisted (chunks). Denoiser became flow.

Spine reminder:

FM = pick a path, regress the VF; OT → straighter/faster
→ DP chunks → π_0 OT-CFM head → BEHAVIOR fixes robot stats

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